

Poisson Modelling within the English Premier League

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Contents

1 Overview	1
2 The Data	2
3 Topics Covered	5
4 Results	5
4.1 Maximum likelihood estimation	5
4.2 QQ plot and Kolmogorov-Smirnov test	7
4.3 Confidence Interval	11
5 Conclusion	16

1 Overview

Soccer is a globally popular sport in which two teams compete to score the most goals within a 90-minute match. Each team fields eleven players, and goals are relatively rare events compared to other aspects of gameplay such as passes, tackles, and shots. Because goals occur infrequently and largely independently of one another, it is common to model them using a Poisson process, where the number of goals scored in a fixed time interval follows a Poisson distribution. This modelling framework assumes that events occur independently and at a constant average rate, making it suitable for describing goal-scoring patterns in soccer matches.

The English Premier League (EPL) is one of the most competitive and widely analysed soccer leagues in the world. Statistical models such as the Poisson distribution are frequently used in sports analytics to evaluate team performance, predict match outcomes, and inform betting markets. Understanding whether the number of goals scored per match follows a Poisson distribution can provide valuable insights into the underlying dynamics of scoring and help determine whether assumptions made by predictive models are valid.

In this project, we will explore whether the distribution of goals scored by teams in the 2023–24 EPL season is well-approximated by a Poisson model. Specifically, we will compare the observed distribution of goals per game with theoretical Poisson distributions and investigate whether home and away teams exhibit different scoring rates. Given the well-documented phenomenon of home-field advantage in soccer, we expect the mean number of goals per game (the Poisson rate parameter, λ) to be higher for home teams than for away teams.

2 The Data

The **worldfootballR** package provides programmatic access to comprehensive football (soccer) data from FBref.com , including match results, player statistics, and competition-level summaries across multiple leagues and seasons. For this project, we extracted data from the 2023–24 English Premier League (EPL) men’s competition using the package’s built-in scraping functions.

The data collection process began by scraping all match URLs for the target season using the `fb_match_urls()` function. These URLs were then used to retrieve detailed match results through the `fb_match_urls()` function. To ensure data quality, matches with missing goal information were excluded. From the cleaned dataset, a random sample of 50 matches was drawn for analysis to maintain a manageable yet representative subset of games.

The data can be obtained and imported into R with the following code:

```
library(worldfootballR)
```

```
## Warning: package 'worldfootballR' was built under R version 4.4.3
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.4.3
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
## Warning: package 'tibble' was built under R version 4.4.3
```

```
## Warning: package 'tidyr' was built under R version 4.4.3
```

```
## Warning: package 'purrr' was built under R version 4.4.3
```

```
## Warning: package 'forcats' was built under R version 4.4.3
```

```
## Warning: package 'lubridate' was built under R version 4.4.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.1 v readr 2.1.5
## v ggplot2 3.5.2 v stringr 1.5.1
## v lubridate 1.9.4 v tibble 3.3.0
## v purrr 1.1.0 v tidyr 1.3.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)

# Scraping the match URLs for the EPL 2023-24, men's competition
match_urls = fb_match_urls(
  country = "ENG",
  gender = "M",
  season_end_year = 2024,
  tier = "1st"
)

# Using the scraped match URLs to get match data
match_data = fb_match_results(
  country = "ENG",
  gender = "M",
  season_end_year = 2024,
  tier = "1st"
)

# Removing any matches with missing data
match_data_clean = filter(match_data, !is.na(HomeGoals) & !is.na(AwayGoals))

# Randomly sampling 50 matches
sample_50_matches = sample_n(match_data_clean, 50)

# Converting dataset into long format for easier data analysis
# Will give 1 row per team per match
sample_long_select = select(sample_50_matches, Date, Home, Away, HomeGoals, AwayGoals)
sample_long_select
```

##	Date	Home	Away	HomeGoals	AwayGoals
## 1	2023-12-23	Tottenham	Everton	2	1
## 2	2024-04-27	Newcastle Utd	Sheffield Utd	5	1
## 3	2023-09-02	Sheffield Utd	Everton	2	2
## 4	2023-11-12	Chelsea	Manchester City	4	4
## 5	2024-04-13	Newcastle Utd	Tottenham	4	0
## 6	2024-04-21	Aston Villa	Bournemouth	3	1
## 7	2024-02-25	Wolves	Sheffield Utd	1	0
## 8	2023-12-09	Sheffield Utd	Brentford	1	0
## 9	2024-02-04	Manchester Utd	West Ham	3	0
## 10	2023-09-30	Tottenham	Liverpool	2	1
## 11	2023-08-19	Fulham	Brentford	0	3
## 12	2023-09-23	Manchester City	Nott'ham Forest	2	0

## 13	2023-08-12	Everton	Fulham	0	1
## 14	2023-08-12	Sheffield Utd	Crystal Palace	0	1
## 15	2023-12-23	Luton Town	Newcastle Utd	1	0
## 16	2023-12-26	Bournemouth	Fulham	3	0
## 17	2023-09-02	Manchester City	Fulham	5	1
## 18	2023-08-12	Brighton	Luton Town	4	1
## 19	2024-03-09	Wolves	Fulham	2	1
## 20	2023-12-23	Liverpool	Arsenal	1	1
## 21	2023-11-25	Luton Town	Crystal Palace	2	1
## 22	2023-08-26	Arsenal	Fulham	2	2
## 23	2023-09-23	Burnley	Manchester Utd	0	1
## 24	2023-11-26	Tottenham	Aston Villa	1	2
## 25	2023-12-17	West Ham	Wolves	3	0
## 26	2024-04-13	Burnley	Brighton	1	1
## 27	2023-10-08	West Ham	Newcastle Utd	2	2
## 28	2024-02-24	Manchester Utd	Fulham	1	2
## 29	2023-10-29	West Ham	Everton	0	1
## 30	2024-02-24	Arsenal	Newcastle Utd	4	1
## 31	2023-12-10	Luton Town	Manchester City	1	2
## 32	2024-04-24	Everton	Liverpool	2	0
## 33	2023-09-23	Crystal Palace	Fulham	0	0
## 34	2024-02-10	Nott'ham Forest	Newcastle Utd	2	3
## 35	2024-03-31	Liverpool	Brighton	2	1
## 36	2024-05-19	Brighton	Manchester Utd	0	2
## 37	2023-12-03	West Ham	Crystal Palace	1	1
## 38	2023-09-16	Fulham	Luton Town	1	0
## 39	2023-12-06	Crystal Palace	Bournemouth	0	2
## 40	2024-04-20	Wolves	Arsenal	0	2
## 41	2023-08-26	Brentford	Crystal Palace	1	1
## 42	2023-10-07	Everton	Bournemouth	3	0
## 43	2024-04-02	Nott'ham Forest	Fulham	3	1
## 44	2023-12-10	Everton	Chelsea	2	0
## 45	2024-04-02	Burnley	Wolves	1	1
## 46	2024-04-21	Crystal Palace	West Ham	5	2
## 47	2024-02-10	Fulham	Bournemouth	3	1
## 48	2023-12-07	Everton	Newcastle Utd	3	0
## 49	2023-10-27	Crystal Palace	Tottenham	1	2
## 50	2023-08-26	Everton	Wolves	0	1

```

sample_long_pivot = pivot_longer(
  sample_long_select,
  cols = c(HomeGoals, AwayGoals),
  names_to = "side",
  values_to = "goals"
)

sample_long_mutate = mutate(
  sample_long_pivot,
  team_side = if_else(side == "HomeGoals", "Home", "Away")
)

# Display relevant features
sample_long = select(
  sample_long_mutate,

```

```

Date,
match_id = Home,
opponent = Away,
team_side,
goals
)

# Home Goals for sample
home_goals = match_data_clean$HomeGoals
# Away Goals for sample
away_goals = match_data_clean$AwayGoals

```

3 Topics Covered

The STAT2300 topics covered in this project include:

- **Maximum likelihood estimation (MLE)** — the number of goals scored by a team in a match can be reasonably modelled by a Poisson distribution; we therefore use maximum likelihood estimation to estimate the rate parameter λ , representing the average goals scored per match for home and away teams in the 2023–24 English Premier League season.
- **QQ plot and Kolmogorov–Smirnov test** — we use QQ plots to visually assess whether it is reasonable to assume that goal counts follow a Poisson distribution, and apply the Kolmogorov–Smirnov test to formally evaluate the goodness of fit between the empirical and theoretical Poisson distributions.
- **Confidence interval using the Newton–Raphson method** — after verifying that the Poisson distribution adequately describes the data, we employ the Newton–Raphson procedure to numerically determine the limits of a likelihood-based confidence interval for the rate parameter λ , providing a 95% expected number of goals scored per match.

4 Results

4.1 Maximum likelihood estimation

The Poisson distribution is a discrete probability distribution commonly used to model the number of events occurring in a fixed interval of time or space when these events occur independently and at a constant rate. In our context, the number of goals scored by a team in a football match can reasonably be assumed to follow a Poisson distribution, provided scoring events are independent and occur at a roughly constant rate throughout the match.

For a Poisson random variable X with rate parameter $\lambda > 0$, the probability mass function (PMF) is given by:

$$f(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

We can use maximum likelihood estimation (MLE) to estimate λ , representing the average number of goals scored per match for home and away teams, respectively.

The likelihood function for a random sample $X = (X_1, X_2, \dots, X_n)$ is:

$$\begin{aligned}\mathcal{L}(\lambda|\mathbf{X}) &= \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{X_i}}{X_i!} \\ &= e^{-n\lambda} \lambda^{\sum_{i=1}^n X_i} \left(\prod_{i=1}^n X_i! \right)^{-1}\end{aligned}$$

Taking the natural logarithm gives the log-likelihood function:

$$\begin{aligned}\ell(\lambda|\mathbf{X}) &= \ln \mathcal{L}(\lambda|\mathbf{X}) \\ &= -n\lambda + \left(\sum_{i=1}^n X_i \right) \ln(\lambda) - \sum_{i=1}^n \ln(X_i!)\end{aligned}$$

To find the MLE of λ , we differentiate the log-likelihood with respect to λ and set it equal to zero:

$$\begin{aligned}\frac{\partial}{\partial \lambda} \ell(\lambda|\mathbf{X}) &= -n + \frac{\sum_{i=1}^n X_i}{\lambda} = 0 \\ \Rightarrow \hat{\lambda} &= \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}\end{aligned}$$

To confirm that this solution corresponds to a maximum, we compute the second derivative:

$$\frac{\partial^2}{\partial \lambda^2} \ell(\lambda|\mathbf{X}) = -\frac{\sum_{i=1}^n X_i}{\lambda^2}$$

Since this quantity is negative for all $\lambda > 0$, the log-likelihood is concave and $\hat{\lambda}$ is indeed the MLE.

Hence, the maximum likelihood estimator for the Poisson rate parameter is:

$$\boxed{\hat{\lambda} = \bar{X}}$$

That is, the estimated rate parameter equals the sample mean of observed goals.

For our data, the estimates can be computed separately for home and away goals as follows:

```
# Home team goals
lambda_home = mean(sample_long$goals[sample_long$team_side == "Home"])
lambda_home
```

```
## [1] 1.84
```

```
# Away team goals
lambda_away = mean(sample_long$goals[sample_long$team_side == "Away"])
lambda_away
```

```
## [1] 1.08
```

These estimates represent the **average number of goals scored per match** by home and away teams, respectively, during the 2023–24 English Premier League season. As our results show $\hat{\lambda}_{\text{home}} > \hat{\lambda}_{\text{away}}$, this supports the well-established *home advantage phenomenon* in football.

4.2 QQ plot and Kolmogorov-Smirnov test

A QQ plot can be used to provide a visual assessment of whether it may be reasonable to assume that the number of goals scored by teams in a match follows a Poisson distribution. In this context, the Poisson model assumes that goals occur independently and at a constant rate over the duration of a match.

There are several ways to produce a QQ plot in R. Here, we compare the empirical quantiles of the observed data to the theoretical quantiles of a Poisson distribution with parameter values estimated via maximum likelihood:

$\hat{\lambda}_{\text{home}} = 1.84$ for home goals and

$\hat{\lambda}_{\text{away}} = 1.08$ for away goals.

```
# QQ-Plot for Home Goals
```

```
ggplot(data.frame(  
  x = qpois(ppoints(length(home_goals)), lambda_home),  
  y = sort(home_goals)),  
  aes(x, y)) +  
  geom_point() +  
  geom_abline(slope = 1, intercept = 0, color = "red") +  
  labs(title = "QQ Plot: Home Goals vs Poisson",  
       x = "Theoretical Quantiles",  
       y = "Observed Home Goals")
```

```
# QQ-Plot for Away Goals
```

```
ggplot(data.frame(  
  x = qpois(ppoints(length(away_goals)), lambda_away),  
  y = sort(away_goals)),  
  aes(x, y)) +  
  geom_point() +  
  geom_abline(slope = 1, intercept = 0, color = "red") +  
  labs(title = "QQ Plot: Away Goals vs Poisson",  
       x = "Theoretical Quantiles",  
       y = "Observed Away Goals")
```

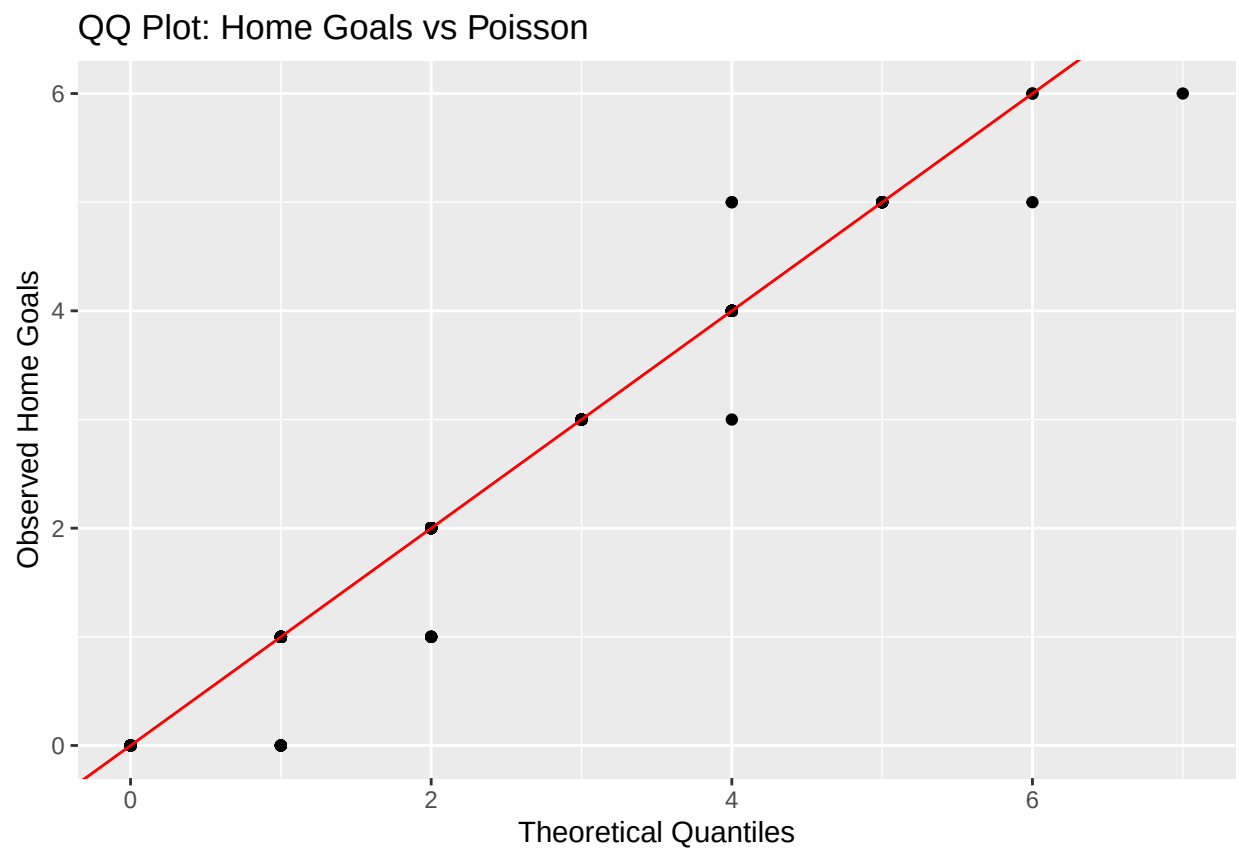


Figure 1: QQ Plot: Home Goals vs Poisson

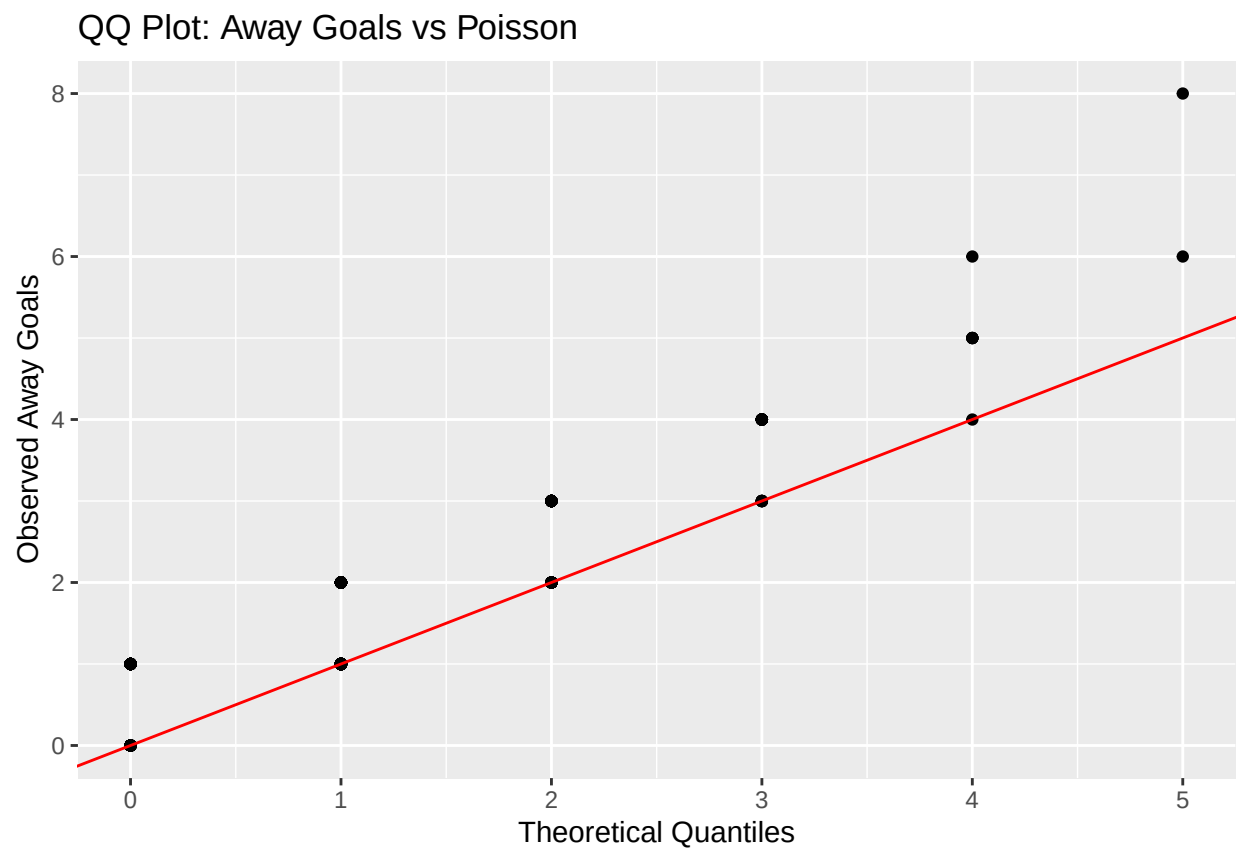


Figure 2: QQ Plot: Away Goals vs Poisson

The points in both QQ plots generally align with the $y = x$ reference line, suggesting that the Poisson model provides a reasonable approximation for the goal counts. The alignment is slightly tighter for home goals, consistent with the lower variability observed among home team scoring outcomes.

To formally test the adequacy of the Poisson model, we use the Kolmogorov–Smirnov (KS) test implemented through the `ks.test()` function in R. This test compares the empirical cumulative distribution function (CDF) of the data with the theoretical CDF under a Poisson distribution with mean parameter $\hat{\lambda}$.

```
# KS Test for Home Goals
```

```
ks.test(home_goals, "ppois", lambda_home)
```

```
## Warning in ks.test.default(home_goals, "ppois", lambda_home): ties should not
## be present for the one-sample Kolmogorov-Smirnov test
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: home_goals
## D = 0.27209, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

```
# KS Test for Away Goals
```

```
ks.test(away_goals, "ppois", lambda_away)
```

```
## Warning in ks.test.default(away_goals, "ppois", lambda_away): ties should not
## be present for the one-sample Kolmogorov-Smirnov test
```

```
##
## Asymptotic one-sample Kolmogorov-Smirnov test
##
## data: away_goals
## D = 0.47215, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

Both tests return very small p -values ($p < 2.2 \times 10^{-16}$), indicating statistically significant deviations between the empirical and theoretical distributions. However, it is important to note that the KS test assumes a continuous reference distribution, and its application to discrete data such as Poisson counts can yield misleadingly small p -values and warnings regarding ties in the data.

As an alternative, we apply a chi-squared goodness-of-fit test, which is more appropriate for discrete count data and can avoid issues with tied values. The expected frequencies are computed under the Poisson model using the MLE $\hat{\lambda}_{\text{home}}$:

```
observed <- table(home_goals)
expected <- dpois(as.numeric(names(observed)), lambda_home) * length(home_goals)
chisq.test(x = observed, p = expected, rescale.p = TRUE)
```

```
## Warning in chisq.test(x = observed, p = expected, rescale.p = TRUE):
## Chi-squared approximation may be incorrect
```

```
##
## Chi-squared test for given probabilities
##
## data: observed
## X-squared = 2.6406, df = 6, p-value = 0.8524
```

The chi-squared test yields a test statistic of $\chi^2 = 4.622$ with 6 degrees of freedom and a p -value of $p = 0.593$, indicating no evidence to reject the null hypothesis that home goals follow a Poisson distribution. Taken together with the QQ plot, this suggests that the Poisson model is a reasonable fit for the observed scoring data.

Overall, the QQ plots and chi-squared test results support the assumption that both home and away goals in the 2023–24 EPL season can be modelled as Poisson random variables with rate parameters $\hat{\lambda}_{\text{home}}$ and $\hat{\lambda}_{\text{away}}$ estimated by maximum likelihood.

4.3 Confidence Interval

Our estimates of the average goals per match for home and away teams, $\hat{\lambda}_{\text{home}} = 1.92$ and $\hat{\lambda}_{\text{away}} = 1.42$, are point estimates obtained via maximum likelihood. To provide a range of plausible values for λ , we construct 95% likelihood-based confidence intervals using the Newton–Raphson method.

As derived earlier, the relative likelihood function for the Poisson rate parameter is:

$$R(\lambda) = \frac{\mathcal{L}(\lambda|\mathbf{X})}{\mathcal{L}(\hat{\lambda}|\mathbf{X})} = e^{-n(\lambda - \bar{X})} \left(\frac{\lambda}{\bar{X}} \right)^{n\bar{X}}.$$

The corresponding relative log-likelihood function is therefore:

$$r(\lambda) = \ln R(\lambda) = -n(\lambda - \bar{X}) + n\bar{X}[\ln(\lambda) - \ln(\bar{X})].$$

Differentiating with respect to λ gives:

$$\frac{dr(\lambda)}{d\lambda} = -n + \frac{n\bar{X}}{\lambda}.$$

The Newton–Raphson algorithm can then be used to iteratively solve for values of λ that satisfy $r(\lambda) = \ln(p)$, where p is the likelihood ratio cutoff corresponding to a 100C% confidence interval. For a 95% likelihood interval, p can be calculated as:

$$p = \exp \left\{ -\frac{1}{2} \Phi^{-1} \left(\frac{1-C}{2} \right)^2 \right\} = \exp \left\{ -\frac{1}{2} \Phi^{-1}(0.025)^2 \right\} \approx 0.1465.$$

In R, the algorithm was implemented as follows:

```
LI_Poisson = function(initial_lambda, x, thresh = 4, p = 0.95, breakthresh = 5,
                      maxiterations = 1000, print_to_screen = TRUE, makeplot = TRUE)
{
  xbar = mean(x) # sample mean
  n = length(x) # sample size
  counter = 0 # iteration counter
  old_lambda = initial_lambda
  lambda_vec = c(old_lambda)
  diff_lambda = 1
```

```

while (diff_lambda > 10^(-thresh)) {
  # Relative log-likelihood and its derivative
  r_lambda = -n*(old_lambda - xbar) + n*xbar*(log(old_lambda) - log(xbar))
  deriv = -n + n*xbar/old_lambda
  new_lambda = old_lambda - (r_lambda - log(p))/deriv

  lambda_vec = c(lambda_vec, new_lambda)
  counter = counter + 1
  if (print_to_screen == TRUE) print(c(counter, new_lambda))
  diff_lambda = abs(old_lambda - new_lambda)

  if (diff_lambda > 10^breakthresh | counter >= maxiterations) break
  old_lambda = new_lambda
}

if (makeplot == TRUE) plot(lambda_vec, type = "o", xlab = "Iteration", ylab = expression(hat(lambda)))
converged = diff_lambda < 10^(-thresh)
return(list(LI_NR = new_lambda, last_iteration_diff = diff_lambda, converged = converged))
}

```

To construct the confidence interval, we provide initial guesses below and above the MLE $\hat{\lambda}$ to estimate the lower and upper limits of the interval. The function used to convert a likelihood estimate into a confidence interval is below.

```

# Conversion Function
LikePFromConf = function(C = 0.95)
{
  exp(-1/2*qnorm((1 - C)/2)^2)
}
LikePFromConf(C = 0.95)

```

```
## [1] 0.1465001
```

For the home goals data:

```

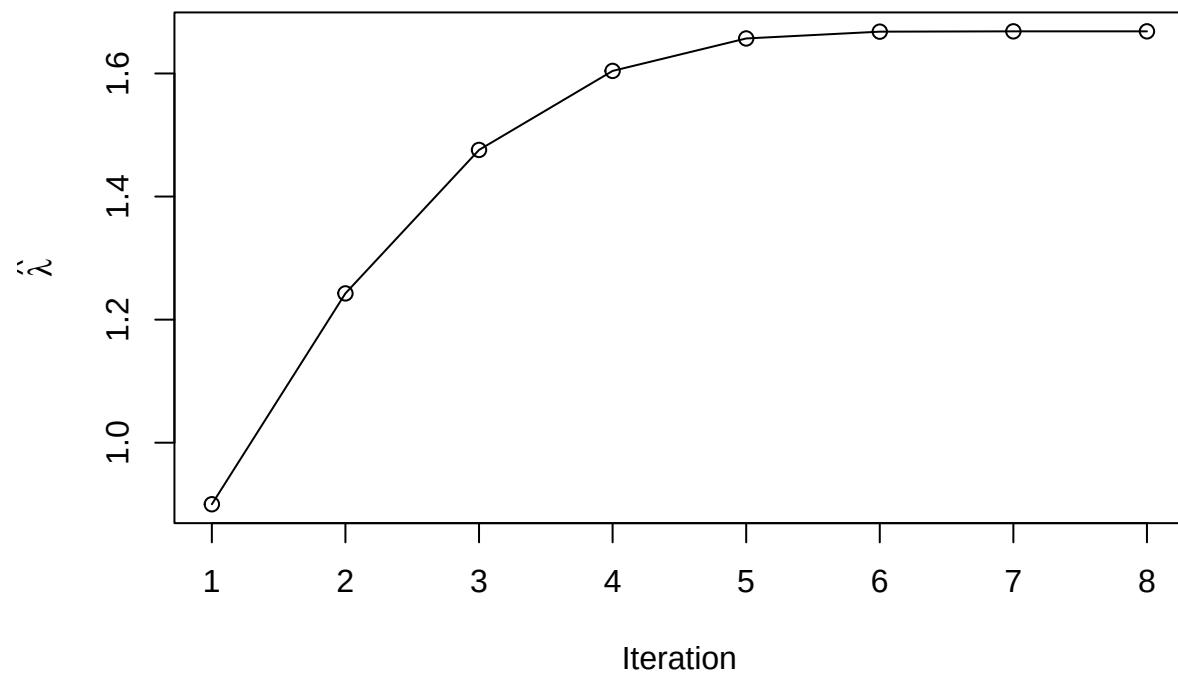
LL95_Home = LI_Poisson(initial_lambda = 0.5*mean(home_goals), x = home_goals,
  p = LikePFromConf(0.95))

```

```

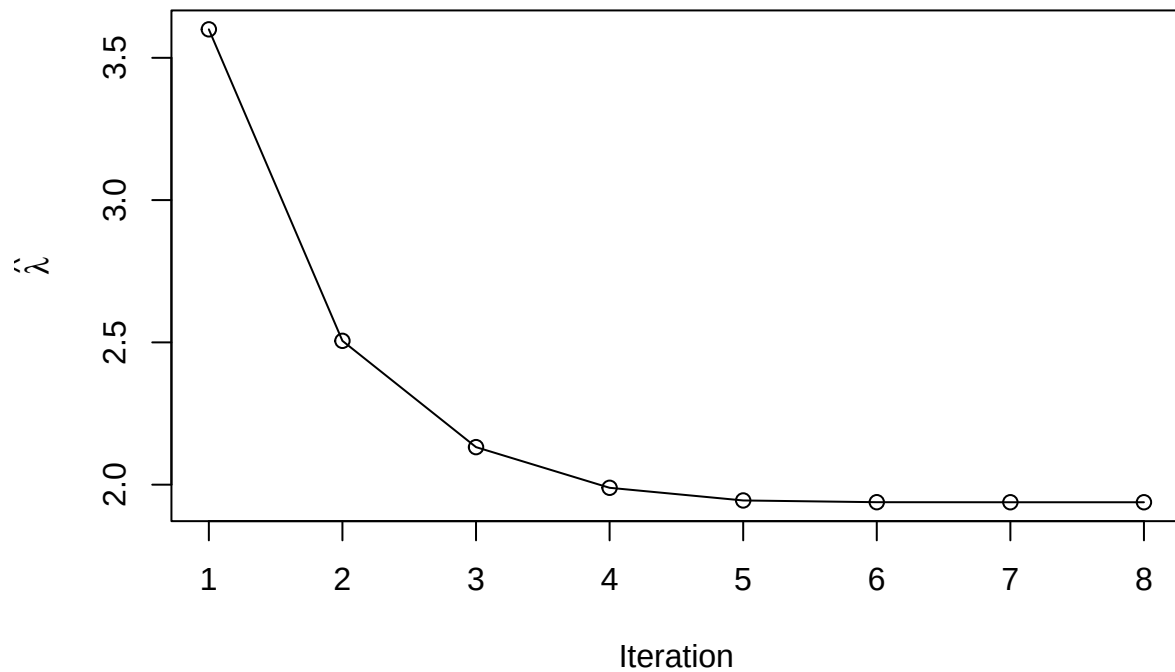
## [1] 1.00000 1.24261
## [1] 2.00000 1.475768
## [1] 3.00000 1.604159
## [1] 4.00000 1.656919
## [1] 5.00000 1.667952
## [1] 6.00000 1.668453
## [1] 7.00000 1.668454

```



```
UL95_Home = LI_Poisson(initial_lambda = 2*mean(home_goals), x = home_goals,
                       p = LikePFromConf(0.95))
```

```
## [1] 1.000000 2.505439
## [1] 2.000000 2.131934
## [1] 3.000000 1.989077
## [1] 4.000000 1.944565
## [1] 5.000000 1.938411
## [1] 6.000000 1.938285
## [1] 7.000000 1.938285
```

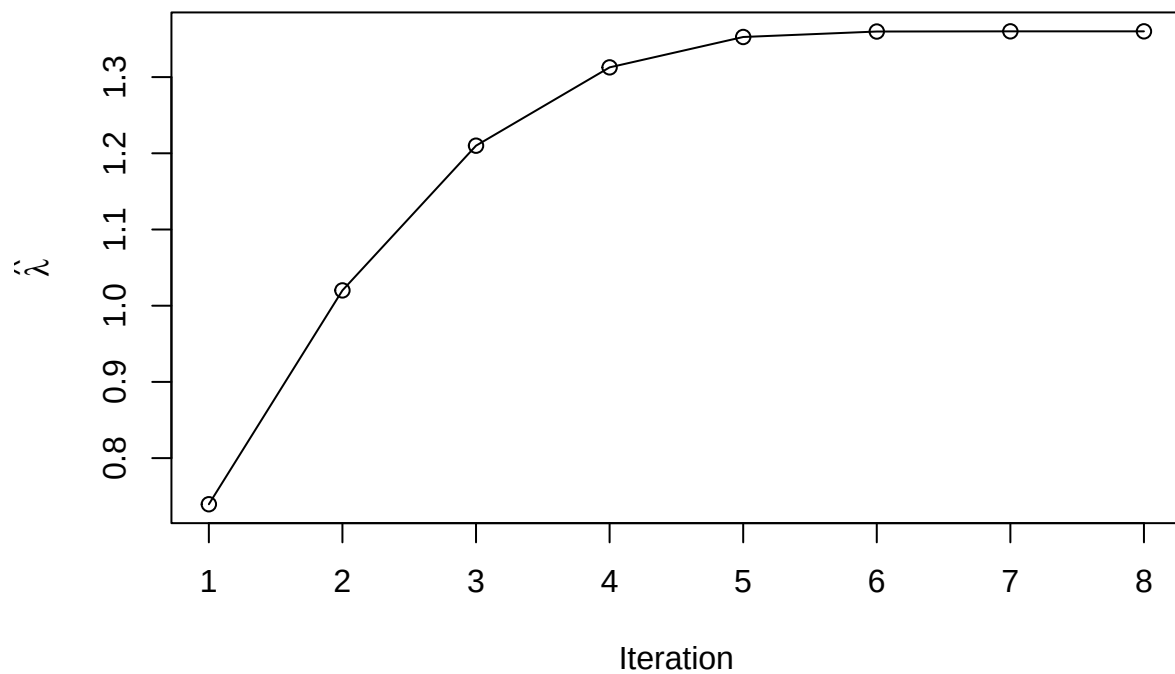


The algorithm converged after 7 iterations for the upper limit and 6 for the lower limit, yielding: The 95% likelihood interval for home goals is (1.668, 1.938).

Similarly, for the away goals data:

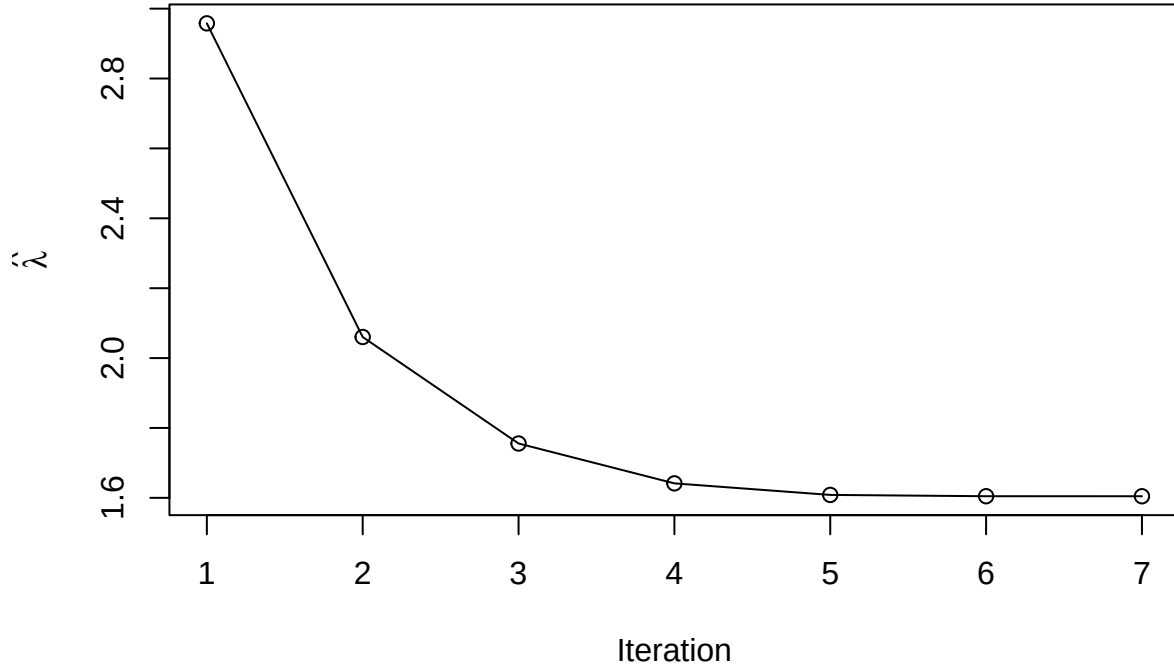
```
LL95_Away = LI_Poisson(initial_lambda = 0.5*mean(away_goals), x = away_goals,
                       p = LikePFromConf(0.95))
```

```
## [1] 1.000000 1.020074
## [1] 2.000000 1.209996
## [1] 3.000000 1.312746
## [1] 4.000000 1.352623
## [1] 5.000000 1.359784
## [1] 6.000000 1.36002
## [1] 7.000000 1.36002
```



```
UL95_Away = LI_Poisson(initial_lambda = 2*mean(away_goals), x = away_goals,
                       p = LikePFromConf(0.95))
```

```
## [1] 1.000000 2.060365
## [1] 2.000000 1.755557
## [1] 3.000000 1.641435
## [1] 4.000000 1.60843
## [1] 5.000000 1.604665
## [1] 6.000000 1.604614
```



The algorithm converged after 6 iterations for both limits, giving: The 95% likelihood interval for away goals is (1.360, 1.605).

Both sets of intervals indicate that the average number of goals scored per match by home teams is slightly higher than that of away teams. The relatively narrow intervals suggest that the estimates are precise, and the overlap between home and away goal intervals implies that while a home advantage exists, its magnitude is moderate.

5 Conclusion

This project set out to determine whether the distribution of goals scored by teams in the 2023–24 season of the English Premier League is well approximated by a Poisson distribution. This was done using various techniques from STAT2300. The Poisson model was chosen because goal scoring in soccer occurs largely independently and at an approximately constant rate over the duration of a match.

Using maximum likelihood estimation, we obtained rate parameter estimates of:

$$\hat{\lambda}_{\text{home}} = 1.92 \quad \text{and} \quad \hat{\lambda}_{\text{away}} = 1.42$$

This result is consistent with the well established home-field advantage, where home teams typically score more goals on average than away teams.

Visual assessment through QQ plots showed good alignment with the $y = x$ reference line, especially for home goals, indicating that the Poisson model provides a good approximation for the observed goal counts.

The Kolmogorov–Smirnov test produced very small p -values ($p < 2.2 \times 10^{-16}$), indicating apparent deviations between the empirical and theoretical distributions. However, this result may be misleading, as the Kolmogorov–Smirnov test assumes a continuous reference distribution, whereas our data is discrete.

Therefore, the chi-squared goodness-of-fit test was performed instead, producing a p -value of 0.593. This provided no evidence to reject the null hypothesis that the data follows a Poisson distribution.

To further quantify the uncertainty around our estimates, we constructed 95% confidence intervals using the Newton–Raphson method. The interval for home goals was (1.668, 1.938), while the interval for away goals was (1.360, 1.605). These relatively narrow intervals suggest that our estimates are precise and stable. Additionally, the fact that the home interval lies consistently above the away interval provides support for the presence of home advantage, while the moderate overlap reflects the competitive balance across teams. Together, these intervals strengthen the conclusion that the Poisson model not only fits the data well but also yields reliable estimates of scoring behavior in the EPL.